

Chapter 7

Unemployment, Inflation, and Long-Run Growth

Lecturer: Mihyun You

Department of Economics
Georgetown University

Summer 2024



GEORGETOWN UNIVERSITY

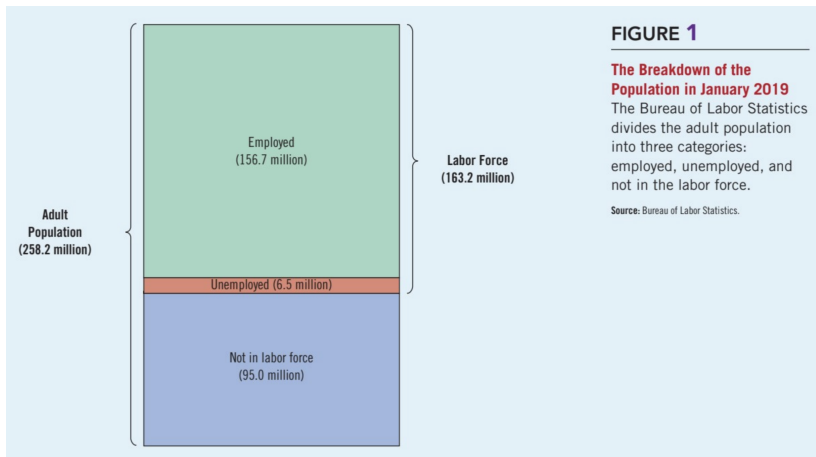
Contents

- ① Unemployment
- ② Inflation
- ③ Long-Run Growth



GEORGETOWN UNIVERSITY

Breakdown of the Population



Identifying Unemployment

- The Current Population Survey (CPS)
 - monthly survey about 60,000 U.S. households conducted by the United States Census Bureau.
- The Bureau of Labor Statistics (BLS) categorizes the population into three categories according to the surveyed answers.
 - ① Employed: Those who *worked* as paid, self-employed, or unpaid workers in a family member's businesses. Both full-time and part-time workers are counted. Those who were temporarily not working due to vacation, illness, or bad weather.
 - ② Unemployed: Those who were not employed, were capable of work, and had tried to find employment during the previous **four weeks**.
 - ③ Not in the labor force: neither employed nor unemployed, such as full-time students, homemakers, **discouraged workers**, and retirees.

Measuring Unemployment

- The working-age population (adult population) is defined as those aged 16 and older.

$$\text{Adult population} = \underbrace{\text{Labor force}}_{\text{Employed} + \text{Unemployed}} + \text{Not in labor force}.$$

- Labor-force participation rate (%) = $\frac{\text{Labor force}}{\text{Adult population}} \times 100$.
- Unemployment rate (%) = $\frac{\text{Unemployed}}{\text{Labor force}} \times 100$.

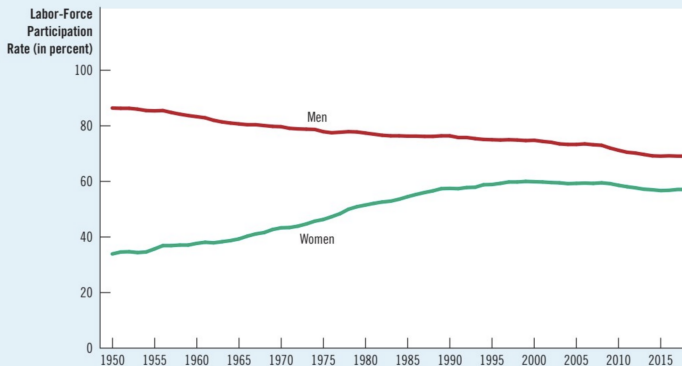
Labor-force Participation Rate by Gender

FIGURE 3

Labor-Force Participation Rates for Men and Women since 1950

This figure shows the percentage of adult men and women who are members of the labor force. It shows that, over the past 60 years, women have entered the labor force and men have left it.

Source: U.S. Department of Labor.



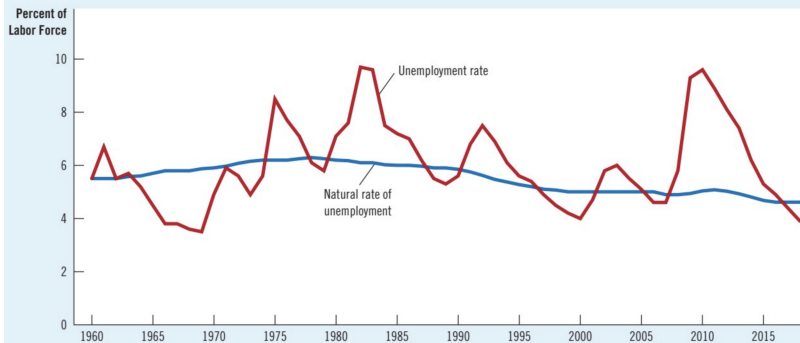
Unemployment Rate

This graph uses annual data on the U.S. unemployment rate to show the percentage of the labor force without a job. The natural rate of unemployment is the normal level of unemployment around which the unemployment rate fluctuates.

Source: U.S. Department of Labor, Congressional Budget Office.

FIGURE 2

Unemployment Rate since 1960



Exercise

End-of-Chapters Problems 1.8

Suppose the number of employed people in an economy is 312,545,372. The unemployment rate in this economy is 7.4 %, or 0.074 , and the labor-force participation rate is 80 %, or 0.8.

- (a) What is the size of the labor force?
- (b) How many people are employed?
- (c) What is the size of the working-age population?

Components of Unemployment

- Natural rate of unemployment: The unemployment rate that occurs as a normal part of the functioning of the economy. The unemployment rate when the economy satisfies the maximum employment and potential level of output.
- Frictional unemployment: short-run job/skill matching problems and switching workplaces.
- Structural unemployment: changes in the structure of the economy that result in a significant loss of jobs in certain industries.
- Cyclical unemployment: unemployment that is above frictional plus structural unemployment.

Discussion Points

Does the measure what we want it to measure?

- Labor market is not static; movements into and out of the labor force are common.
- The official unemployment rate is useful but imperfect measure of joblessness.
- The unemployment rate measure underestimates the fraction of people who are involuntarily out of work.
- ★ Discouraged-worker effect
- The unemployment rate measure does not have something to say about the *quality* of the job.

Complementary Measures to Unemployment

Measures of Labor Underutilization

Measure and Description		Rate
U-1	Persons unemployed 15 weeks or longer, as a percent of the civilian labor force (includes only very long-term unemployed)	1.3%
U-2	Job losers and persons who have completed temporary jobs, as a percent of the civilian labor force (excludes job leavers)	1.9
U-3	Total unemployed, as a percent of the civilian labor force (official unemployment rate)	4.0
U-4	Total unemployed, plus discouraged workers, as a percent of the civilian labor force plus discouraged workers	4.3
U-5	Total unemployed plus all marginally attached workers, as a percent of the civilian labor force plus all marginally attached workers	4.9
U-6	Total unemployed, plus all marginally attached workers, plus total employed part-time for economic reasons, as a percent of the civilian labor force plus all marginally attached workers	8.1

Note: The Bureau of Labor Statistics defines terms as follows.

- *Marginally attached workers* are persons who currently are neither working nor looking for work but indicate that they want and are available for a job and have looked for work sometime in the recent past.
- *Discouraged workers* are marginally attached workers who have given a job-market-related reason for not currently looking for a job.
- *Persons employed part-time for economic reasons* are those who want and are available for full-time work but have had to settle for a part-time schedule.

TABLE 2

Measures of Labor Underutilization

The table shows various measures of joblessness for the U.S. economy. The data are for January 2019.

Source: U.S. Department of Labor.

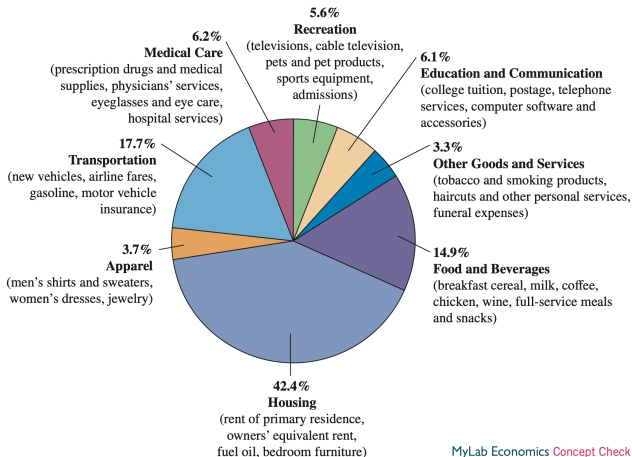
Measuring the Cost of Living

- Macroeconomists are concerned with changes in the *overall* price level of goods and services.
- However, all prices for the multitude of goods and services do not move together at the same rate, and this fact makes measurement of inflation difficult.
 - Various inflation measures and different calculation method: GDP deflator, CPI, PPI (producer price index), ...

The Consumer Price Index (CPI)

- CPI: A measure of the overall cost of the goods and services bought by a typical consumer
- First constructed during World War I as a basis for adjusting shipbuilders' wages, which the government controlled during the war.
- The BLS compute the CPI each month using a bundle of goods meant to represent the "*market basket*" purchased by the typical consumer.

The CPI Market Basket



MyLab Economics Concept Check

▲ FIGURE 7.1 The CPI Market Basket

The CPI market basket shows how a typical consumer divides his or her money among various goods and services. Most of a consumer's money goes toward housing, transportation, and food and beverages.

How to Calculate CPI

- 1 Step 1: Survey consumers to determine a fixed basket of goods.

Basket = 4 hot dogs, 2 hamburgers

- 2 Step 2: Find the price of each good in each year.

Year	Price of Hot Dogs	Price of Hamburgers
2022	\$1	\$2
2023	2	3
2024	3	4

- 3 Step 3: Compute the cost of the basket of goods in current year

How to Calculate CPI

- ④ Step 4: Choose one year as a base year (2022) and compute the CPI in each year

$$\text{CPI} = \frac{\text{Price of basket in current year}}{\text{Price of basket in base year}} \times 100.$$

- ⑤ Step 5: Use the CPI to compute the inflation rate from previous year

$$\text{Inflation rate in year } t = \frac{\text{CPI}_t - \text{CPI}_{t-1}}{\text{CPI}_{t-1}} \times 100.$$

Discussion Points

Limitations of CPI

- Substitution bias
 - A fixed basket ignores the possibility of consumer substitution and, therefore, overstates the increase in the cost of living.
- Introduction of new goods: benefit of diversity
 - A fixed basket does not reflect the increase in value of the dollar resulting from the introduction of new goods.
- Unmeasured quality change
 - If the quality of the goods contained in a fixed basket rises from one year to the next, the value of a dollar rises.

The Costs of Inflation

Correcting economic variables for the effects of inflation

- Indexation
 - Inflation deteriorates your ability to purchase goods and services.
 - COLAs(cost-of-living adjustments): Long-term contracts, social security benefits, pension plans, and wages are often indexed in order to compensate for the costs of inflation.
- Real Interest Rates
 - Inflation affects the income distribution.
 - Debtors are better off and creditors are worse off during an inflation.

Real interest rate = Nominal interest rate – Inflation rate.

- The costs of anticipated inflation can be circumvented, but still, unanticipated inflation may have large effects on the economy.

Looking across Two Millennia

Global GDP over the long run

Our World
in Data

Total output of the world economy. These historical estimates of GDP are adjusted for inflation. We combine three sources to create this time series: the Maddison Database (before 1820), the Maddison Project Database (1820–1989), and the World Bank (1990 onward).



Data source: World Bank (2023); Bolt and van Zanden - Maddison Project Database 2023; Maddison Database 2010

Note: This data is expressed in international-\$¹ at 2017 prices.

OurWorldInData.org/economic-growth | CC BY

Economic Growth as Quite Recent Phenomenon

- From a broader historical perspective—say, the past 2000 years or so—growth rates achieved by rich economies since 1950 seem exceptionally high.
- Moreover, the historical record seems more accurately described by leapfrogging rather than convergence.
- From the end of the Roman Empire (476-1500), there was essentially no growth of output per person in Europe.
- This period of stagnation of output per person is often called the *Malthusian era*.
- On the scale of human history, therefore, sustained growth of output per person is definitely a recent phenomenon.

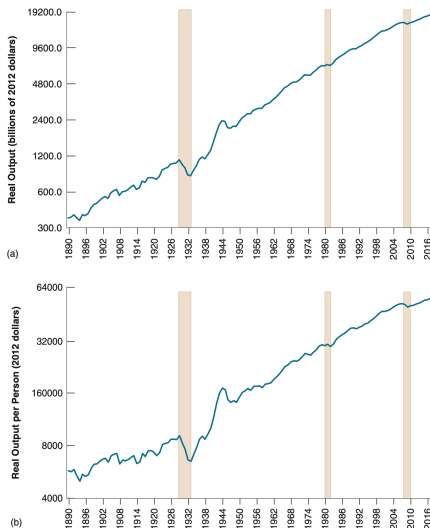
Economic Growth of US since 1890

Figure 10-1

**US GDP since 1890 and
US GDP per Person since
1890**

Panel (a) shows the enormous increase in US output since 1890, by a factor of 51. Panel (b) shows that the increase in output is not simply the result of the large increase in US population from 63 million to more than 320 million over this period. Output per person has risen by a factor of 10.

Sources: 1890–1947: *Historical Statistics of the United States*. <http://hsus.cambridge.org/>; HSUSWeb/toc/hsus-home.do. 1948–2014: *National Income and Product Accounts: Population estimates 1890–2014*: Louis Johnston and Samuel H. Williamson, “What Was the US GDP Then?” Measuring Worth, 2015, www.measuringworth.com/datasets/usgdp/. More recent data: FRED: GDPC1, B230RC0Q173SBEA.



Growth across Rich Countries since 1950

What facts about growth can we infer from the following data?

Table 10-1 The Evolution of Output per Person in Four Rich Countries since 1950				
	Annual Growth Rate Output per Person (%)	Real Output per Person (2011 dollars)		
	1950–2017	1950	2017	2017/1950
France	2.6	7,025	39,461	5.6
Japan	4.1	2,531	40,374	15.9
United Kingdom	2.1	9,354	39,128	4.2
United States	2.0	14,569	54,995	3.8
Average	2.7	8,370	43,490	5.2
The average in the last line is a simple unweighted average.				
Source: Penn World Table Version 8.1./Feenstra, Robert C., Robert Inklaar and Marcel P. Timmer (2015), "The Next Generation of the Penn World Table" forthcoming American Economic Review.				

Determinants of Economic Growth

- Physical capital
 - Human capital
 - Natural resources
 - Technological knowledge
-
- Population growth

Convergence versus Divergence

- If two countries have the same economic conditions, will they experience the same economic growth?
- Institutional factors also matter.
 - Political stability
 - Property rights
 - Free trade